WorkForceX

AI Workforce Transformation & Reskilling Intelligence

ROUND 2 — APPROACH NOTE

Team: MIND SPARK

An evidence-grounded workforce intelligence platform for understanding skill demand, identifying capability gaps, and supporting role transitions and reskilling decisions.

WorkForceX APPROACH

Market Demand

↓

Capability Analysis

↓

Skill Gap Identification

↓

Role Transition

↓

Workforce Decision Support

Evidence Base

Analytics Jobs • DataScience Jobs • JDS Skill Traits • SDS Personality Traits

Analysis based on hackathon-supplied datasets and 2024–25 sample data.

Understand the Change → Model the Transition → Build the Workforce Ahead

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1. Executive Summary

WorkforceX is proposed as an analytics and decision-support system for understanding data-science and analytics workforce requirements. The official hackathon context asks teams to identify meaningful questions from datasets covering job postings, technical skill traits of junior data scientists, and personality traits of senior/customer-facing data scientists. The solution developed by Team Mind Spark uses the four supplied datasets as separate but complementary evidence layers rather than attempting an unsupported row-level merge.

The market layer uses Analytics Jobs and DataScience Jobs to understand role demand, skills, salary bands, experience requirements, company activity and geographic concentration. The talent-outcome layer uses JDS Skill Traits to examine whether measured technical capabilities are associated with a high or low performance-based salary hike. The senior-talent layer uses SDS Personality Traits to examine whether Big Five personality measurements are associated with high or low success classification.

The supplied data contain 15,841 Analytics Jobs records, 1,602 Data Science Jobs records, 139 junior-data-scientist records and 161 senior-data-scientist records. The analysis found no exact duplicate rows in any of the four files. The Analytics Jobs file contains substantial missingness in job description and job type fields, while the other three datasets are structurally complete.

For the predictive component, two transparent classification baselines were evaluated: Logistic Regression and Random Forest. Five-fold stratified cross-validation was used because the outcome datasets are small. On the JDS dataset, Random Forest achieved mean accuracy of 84.1% and mean F1 of 84.5%, while Logistic Regression provided a strong ROC-AUC of 90.3%. On the SDS dataset, Random Forest achieved mean accuracy of 95.0%, F1 of 95.2% and ROC-AUC of 99.2%.

The results are treated as exploratory evidence, not as causal proof or production-grade prediction. The main contribution of WorkforceX is the connected analytical workflow: market demand → capability evidence → outcome patterns → workforce-development implications. This structure directly addresses the hackathon evaluation emphasis on problem identification, data management, analytical skills, conclusions and implications.

2. Problem Definition

The central workforce problem is that organizations may observe changing demand for data-science and analytics capabilities without having an integrated analytical view of what the market is asking for, which technical capabilities are associated with positive junior-career outcomes, and which personality patterns are associated with senior-level success classifications. Looking only at job counts is insufficient; looking only at skills is also insufficient. A useful workforce analytics approach should connect market signals with talent characteristics and observed outcomes.

The official hackathon brief defines the context around data-science-related jobs, skills and personality traits. It explicitly encourages analyses involving data management, visualizations, patterns, statistical or data-mining analysis with a definitive objective to bring out useful information from the available data. Therefore, the problem definition for WorkforceX is intentionally evidence-led and limited to the four datasets supplied by the organizers.

WorkforceX addresses three linked decision questions. First, what does the market currently demand in terms of roles, skills, experience, salary and geography? Second, within the supplied junior

data-scientist sample, which technical skill dimensions are associated with high versus low salary-hike outcomes? Third, within the supplied senior/customer-facing sample, which personality dimensions are associated with high versus low success classifications?

The problem is therefore not framed as predicting an individual's future with certainty. Instead, it is framed as identifying patterns that can support workforce development, training prioritization, career planning and organizational capability discussions. This distinction is important because the supplied outcome datasets are small, masked/sample data and the official instructions warn that the data may contain meaningless, misspelled or mistyped entries and outliers.

WorkforceX consequently adopts a decision-support philosophy. The system should surface evidence, explain the basis of an analytical result and leave final workforce decisions to authorized human decision-makers. This also reduces the risk of turning an exploratory model into an unjustified automated employment decision.

3. Analytics Objective

Primary analytics objective: to build an evidence-grounded workforce intelligence workflow that describes analytics and data-science job-market demand, identifies important technical skill patterns associated with junior salary-hike outcomes, identifies personality-trait patterns associated with senior success classifications, and converts those findings into actionable workforce-development insights.

Objective A — Market intelligence. Quantify role demand, skill demand, experience requirements, salary distributions, company activity and geographic concentration using the Analytics Jobs and DataScience Jobs files. This provides the external-demand side of the WorkforceX decision framework.

Objective B — Junior capability analysis. Determine whether measured technical skill scores differ between junior data scientists classified as high versus low on performance-based salary hike. Evaluate interpretable classification models and identify the skill dimensions contributing most to classification performance.

Objective C — Senior success analysis. Determine whether normalized Big Five personality measurements differ between senior/customer-facing data scientists classified as high versus low success. Evaluate classification models and identify the personality dimensions most important to model discrimination.

Objective D — Consolidation for decision support. Create a common analytical representation across the datasets without forcing an invalid row-level join. The datasets are consolidated at the evidence layer: market demand, junior technical capability patterns and senior personality patterns become complementary inputs to WorkforceX.

Objective E — Prototype translation. Connect the analytical findings to the WorkforceX interface: Executive Dashboard, Workforce X-Ray, Skill Transformation Map, Transition Graph and Role Intelligence. The current prototype is treated as a decision-support visualization layer; illustrative employee-level screens are not claimed to be real company records.

Success criteria for Phase 1 are therefore: clean and documented data; reproducible derivations; meaningful descriptive and inferential analysis; transparent model evaluation; defensible

conclusions; and clear implications for organizations, employees and workforce-development stakeholders.

4. Scope, Stakeholders and Analytical Boundaries

The primary stakeholder is an organizational decision-maker such as an HR, talent, learning and development, workforce-planning or business leader. The Executive Dashboard is therefore designed as an organizational view rather than an individual employee dashboard. A secondary stakeholder is the employee or learner, for whom the system can eventually translate capability gaps into development pathways.

The scope includes the four official datasets only. The Analytics Jobs file represents approximately 15.8 thousand analytics-area job postings; DataScience Jobs contains approximately 1.6 thousand records describing company/job activity; JDS Skill Traits contains approximately 140 junior data-scientist records; and SDS Personality Traits contains approximately 160 senior/customer-facing data-scientist records. The official data description identifies the years for the job datasets as 2024–25.

The scope excludes claims about causality, guaranteed promotion, guaranteed salary increases, guaranteed career transitions and general population personality effects. The models estimate classification patterns in the supplied samples. They are useful for analytical exploration and prototype decision support, but they require larger, representative and longitudinal data before operational deployment.

The market datasets also have different units of observation. Analytics Jobs is a job-posting-level dataset, whereas DataScience Jobs contains company/job records with a field for number of postings. JDS and SDS are person-level sample datasets. Because these units differ, consolidation is performed conceptually and analytically rather than through a single master table.

This boundary is a strength rather than a weakness. It prevents false precision and ensures that each dataset is used for the question it can legitimately answer.

5. Data Sources and Dataset Architecture

Dataset	Rows	Primary unit	Key variables	Analytical role
Analytics Jobs	15,841	Job posting	Experience, description, designation, job type, skills, location, salary	Market and skill-demand analysis
DataScience Jobs	1,602	Company/job record	Company, job title, experience, salary range, number of jobs	Data-science demand and compensation analysis
JDS Skill Traits	139	Junior data scientist	5 technical skill scores + salary-hike outcome	Technical skill/outcome modelling
SDS Personality Traits	161	Senior/customer-facing data scientist	5 personality traits + success outcome	Personality/outcome modelling

The architecture is intentionally modular. Analytics Jobs and DataScience Jobs answer market questions; JDS and SDS answer talent-outcome questions. The four datasets are not assumed to represent the same people, organizations or time series. The analytical consolidation therefore takes place after dataset-specific preparation.

The official data dictionary describes JDS technical scores as averages summarized on a 1–5 scale based on tests, training, evaluations and feedback. The SDS file is based on the Big Five personality framework and uses normalized scores. The target variables are binary: 1 indicates high salary hike for JDS and 1 indicates high success classification for SDS.

6. Overall Approach Description

The WorkforceX analytical pipeline follows a sequence designed to satisfy the hackathon's emphasis on data management, data manipulation, data derivation, consolidation, preparation, analysis, conclusions and implications.

1. Acquire and profile all four official files.
2. Check data types, missing values, duplicates, cardinality, invalid or inconsistent formats and target balance.
3. Derive standardized analytical variables such as experience midpoint and salary midpoint from the Analytics Jobs file.
4. Normalize skill text for frequency analysis by trimming whitespace, removing trailing ellipses and applying case normalization.
5. Parse DataScience Jobs salary strings into numeric lakh-based fields for descriptive comparison.
6. Perform descriptive analysis using counts, percentages, means, medians, standard deviations and cross-tabulations.
7. Compare junior skill scores across high/low salary-hike groups and senior personality scores across high/low success groups.
8. Train interpretable baseline and tree-based classification models using stratified five-fold cross-validation.
9. Identify model feature importance and connect results to workforce-development implications.
10. Translate analytical findings into the WorkforceX prototype while labeling illustrative interface values separately from dataset-derived results.

The design intentionally separates descriptive evidence from predictive modelling. A chart showing skill frequency is not treated as a prediction; a model probability is not treated as a causal statement. This separation is central to a credible analytics report.

The current prototype also contains a role-transition concept based on skill overlap. Because the supplied datasets do not contain real employee-to-future-role transition histories, the transition graph is treated as a prototype decision-support mechanism rather than a validated transition predictor.

7. Data Exploration: Initial Profiling

The Analytics Jobs dataset contains 15,841 records and 8 original columns. It has no exact duplicate rows. The job description field has 3,508 missing records (22.1%), and job type has 12,011 missing records (75.8%). The key-skills field has only 1 missing value. The file contains 10,097 unique designations and 1,355 unique location strings.

The DataScience Jobs dataset contains 1,602 records, no exact duplicates and no missing values across its original eight fields. The JDS and SDS datasets also contain no missing values and no exact duplicates. This completeness makes the two outcome datasets directly usable after feature selection and target verification.

The official hackathon instructions warn that the data may contain meaningless, misspelled or mistyped entries and outliers. Therefore, absence of missing values is not treated as proof of perfect quality. Text standardization, range checks and distribution inspection are still necessary.

Target balance is reasonably close to even. In JDS, 52.5% of records are classified as high salary hike and 47.5% as low. In SDS, 52.8% are classified as high success and 47.2% as low. This reduces the immediate need for class rebalancing, although cross-validation should remain stratified.

The initial profiling stage therefore indicates that the main data-preparation challenge is not missingness in the modelling datasets; it is semantic cleaning and derivation in the job-market datasets, plus careful interpretation of the small outcome samples.

8. Data Manipulation

Data manipulation was performed to convert source formats into analytically useful structures without changing the meaning of the original variables.

Experience manipulation: the Analytics Jobs experience field contains text such as '6-10 yrs'. The lower and upper numeric values were extracted into minimum and maximum experience fields. A midpoint was then derived as: Experience Midpoint = (Minimum Experience + Maximum Experience) / 2. This allows salary-band comparisons and descriptive statistics without pretending that the original source provided a single exact experience requirement.

Salary manipulation: the Analytics Jobs salary field contains six categorical bands such as 0to3, 3to6 and 25to50. For exploratory numeric comparisons, representative midpoints were assigned: 0to3 → 1.5, 3to6 → 4.5, 6to10 → 8, 10to15 → 12.5, 15to25 → 20 and 25to50 → 37.5. These are analytical midpoints, not reported salaries, and are explicitly treated as derived approximations.

DataScience Jobs salary manipulation: values such as 9.7L were converted to numeric lakh values. This creates comparable min, average and max salary fields for aggregation. The number-of-jobs variable was retained as a demand measure and summed by job title or company when the business question required aggregated posting volume.

Text manipulation: skill strings were split on commas, converted to lowercase, trimmed and cleaned of trailing ellipses. This step improves frequency counting while preserving the original source for auditability. The approach is deliberately conservative; it does not attempt aggressive natural-language semantic merging of every synonym.

Column names in the SDS file were also standardized in the analytical layer, including removal of accidental leading/trailing spaces in names such as extraversion. Original values were not altered.

9. Data Derivation

Derived variables are used to make the data suitable for analysis. Each derivation has a transparent formula and a clear business interpretation.

Derived variable	Formula / rule	Purpose
Experience minimum	First numeric value in experience range	Lower-bound requirement
Experience maximum	Second numeric value in experience range	Upper-bound requirement
Experience midpoint	(Min + Max) / 2	Comparable experience indicator
Salary midpoint	Midpoint of source salary band	Exploratory numeric salary indicator
Skill frequency	Count of normalized skill mentions	Market skill demand signal
Aggregated DS postings	Sum(num_of_jobs) by role/company	Demand volume signal
Target balance	Count(target class) / total records	Model evaluation check

The derived fields were designed to support descriptive analysis, not to create artificial precision. For example, a salary midpoint of 12.5 for the 10to15 band does not mean every posting paid 12.5 lakh. It is only a representative value for comparing the broad salary bands.

For JDS and SDS, the feature values are already numeric and therefore require less derivation. The main preparation task is ensuring that the five predictors are treated as continuous scores and the target is treated as a binary classification variable.

Feature importance is another derived analytical output. Random Forest importance is based on impurity reduction within the fitted ensemble. Because this measure can be biased by feature structure and does not establish causality, it is reported as a model-specific importance signal rather than a universal ranking of human capability.

10. Data Consolidation

The four files cannot be safely concatenated row-by-row because they describe different entities and do not share a common employee identifier. Attempting to join them directly would create false relationships. Instead, WorkforceX consolidates them through a common decision framework.

Evidence layer	Dataset	Question answered	Output
Market demand	Analytics Jobs	What skills, roles, locations and salary bands appear in analytics postings?	Skill and role demand signals
Data-science demand	DataScience Jobs	Which data-science roles and companies show posting activity and what compensation is recorded?	Role/company demand and salary signals
Junior talent	JDS Skill Traits	Which technical skill patterns are associated with high/low salary-hike classification?	Skill-outcome patterns and classifier
Senior talent	SDS Personality Traits	Which personality patterns are associated with high/low success classification?	Trait-outcome patterns and classifier

The consolidation therefore happens at the analytical layer: market signals provide external demand context; junior skill analysis provides capability-development evidence; senior trait analysis provides an additional talent-pattern perspective. The final WorkforceX recommendation layer can reference all three without claiming that the same individuals appear in each dataset.

This design also makes the system extensible. If a future phase supplies an organization's actual employee skill inventory, a role-skill mapping can be added as a separate internal workforce layer.

11. Data Preparation and Quality Controls

Preparation begins with schema validation. Numeric fields are checked for numeric types, categorical fields for expected categories, salary fields for parseable numeric values, and experience fields for non-negative ranges. The Analytics Jobs experience data range from 0 to 23 years at the minimum and 0 to 30 years at the maximum after parsing. These values are plausible enough for exploratory use but still require contextual interpretation.

Missingness is handled according to analytical purpose. The 3,508 missing job descriptions are not imputed with invented text. Instead, the description field is excluded from analyses that require textual content unless a separate text-processing strategy is introduced. The 12,011 missing job-type values are retained as missing rather than falsely classified.

Duplicates were checked using full-row equality. No exact duplicate rows were found in any of the four supplied datasets. This does not eliminate the possibility of near-duplicates or repeated postings with different identifiers; therefore, the report avoids claiming that the market datasets contain no duplicate job advertisements.

Outlier treatment is conservative. In job-market data, unusually high salaries, experience requirements or posting counts may represent legitimate observations. Instead of automatically deleting them, the analysis uses distributions, medians and grouped comparisons and flags extreme values for future review.

For model preparation, the JDS and SDS target classes are sufficiently balanced for standard binary classification. Features are numeric and require no one-hot encoding. Standardization is applied inside the Logistic Regression pipeline so that scaling is learned only from the training folds during cross-validation. Random Forest does not require scaling.

These controls make the analytical workflow reproducible and reduce the risk that preprocessing decisions leak information across validation folds.

12. Data Analysis: Analytics Jobs Market

The Analytics Jobs dataset contains 15,841 postings. The mean derived experience midpoint is 6.19 years. The dataset contains 10,097 unique job-designation strings, indicating substantial role diversity. The six salary bands are unevenly distributed, with the 10to15 band representing 22.8% of postings and the 15to25 band representing 20.7%.

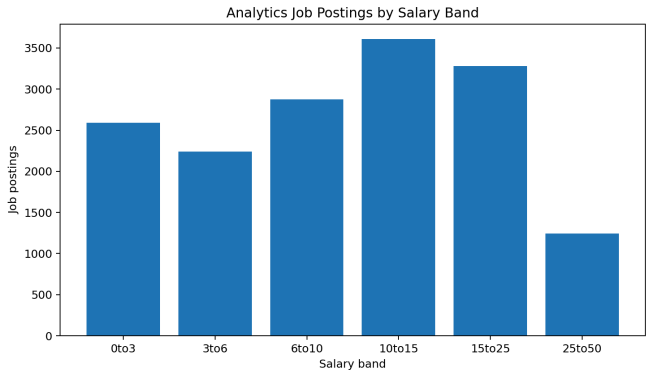


Figure 1. Distribution of Analytics Jobs postings across source salary bands.

The salary distribution suggests that the market sample is concentrated in the middle salary bands rather than the highest band. The 25to50 category represents only 7.9% of postings. This should not be interpreted as a complete labor-market salary distribution because the source is a sample of postings and the salary field is categorical.

A notable analytical relationship appears between experience and salary band. The mean experience midpoint rises monotonically from the lowest salary band to the highest. The correlation between the derived experience midpoint and the representative salary midpoint is approximately 0.66, indicating a moderate-to-strong positive association in this sample.

13. Data Analysis: Skill Demand and Geographic Signals

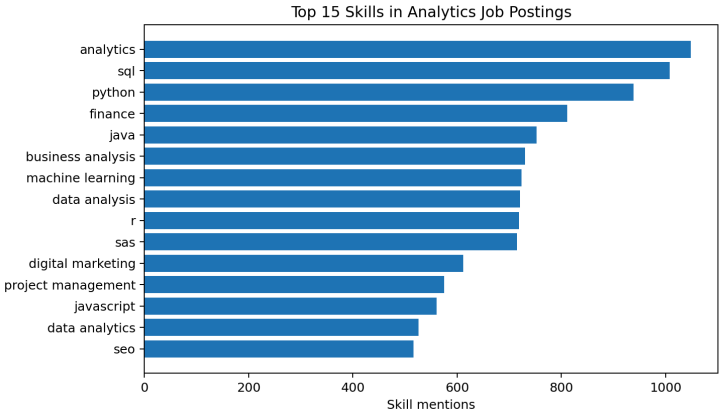


Figure 2. Top 15 normalized skill mentions in Analytics Jobs.

The most frequently observed normalized skill mentions are analytics, SQL, Python, finance, Java, business analysis, machine learning, data analysis, R and SAS. SQL appears in 1,008 skill mentions and Python in 938. These frequencies should be read as posting-level demand signals rather than percentages of the entire labor market because one job posting can contain multiple skills.

The skill results show that the market is heterogeneous. Technical skills such as SQL, Python, Java, machine learning, R and SAS coexist with business and domain skills such as finance, business analysis, project management and digital marketing. This supports the WorkforceX idea that workforce transformation should not be reduced to a single technology skill.

Geographic concentration is also visible. After splitting multi-location strings and normalizing Gurgaon/Gurugram, the most frequently mentioned locations include Bengaluru, Mumbai, Gurgaon/Gurugram, Delhi NCR, Pune, Hyderabad and Chennai. These are posting-location signals rather than counts of unique employers.

From a workforce-development perspective, the combined skill and geography view suggests that capability planning should consider both technical depth and the business context in which analytics work is performed.

14. Data Analysis: Data Science Jobs

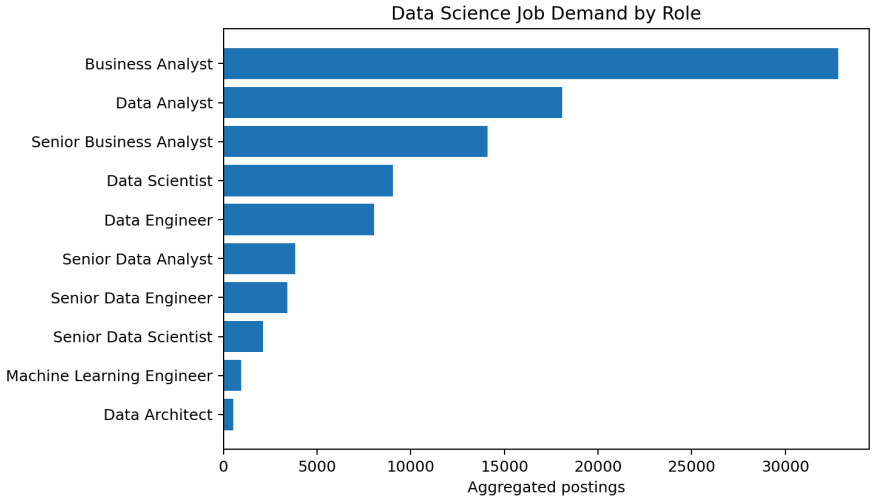


Figure 3. Aggregated number of postings by Data Science Jobs role.

The DataScience Jobs file contains 1,602 company/job records with an aggregated 93,005 postings represented by the num_of_jobs field. Business Analyst has the highest aggregated posting count at 32,843, followed by Data Analyst at 18,095 and Senior Business Analyst at 14,115.

Among the specifically data-science-oriented roles, Data Scientist records 9,051 aggregated postings, while Senior Data Scientist records 2,129. Data Engineer records 8,044 and Senior Data Engineer 3,411. Data Architect has a smaller posting volume but a high mean recorded average salary in the sample.

The salary analysis shows a clear seniority pattern in the supplied sample. The mean recorded average salary is approximately 22.29 lakh for Senior Data Scientist and 18.998 lakh for Senior Data Engineer, compared with approximately 13.53 lakh for Data Scientist and 11.81 lakh for Data Engineer. Data Architect has the highest mean recorded average salary at approximately 25.09 lakh, alongside the highest average minimum experience requirement among the listed roles.

These values are descriptive and should not be generalized as national salary benchmarks. Their primary value for WorkforceX is to establish the relationship between role demand, experience and compensation within the supplied sample.

15. Data Analysis: Junior Data Scientist Skill Outcomes

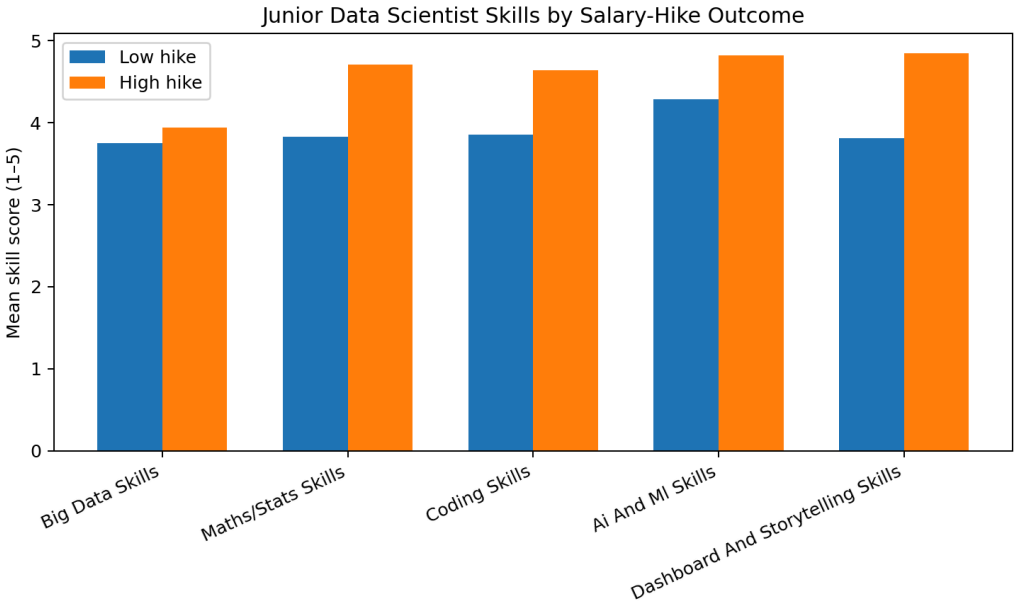


Figure 4. Mean technical skill scores by salary-hike classification in JDS.

The JDS dataset contains 139 junior data-scientist records. The high salary-hike group contains 52.5% of observations. Across all five technical dimensions, the high-hike group has a higher mean score than the low-hike group.

Skill dimension	Low hike mean	High hike mean	Difference
big data skills	3.75	3.94	0.19
maths/stats skills	3.83	4.71	0.88
coding skills	3.85	4.64	0.79
ai and ml skills	4.28	4.82	0.54
dashboard and storytelling skills	3.81	4.85	1.03

The largest observed mean differences are in dashboard and storytelling skills (difference ≈ 1.03 points), mathematics/statistics (≈ 0.88), coding (≈ 0.79) and AI/ML (≈ 0.54). Big data skills show the smallest difference (≈ 0.19). Independent t-tests also show very small p-values for mathematics/statistics, coding, AI/ML and dashboard/storytelling, while the difference in big-data skills is not statistically significant at conventional levels.

These tests are exploratory and should not be interpreted as proving that increasing a specific skill causes a salary hike. The outcome may reflect many omitted factors such as role scope, tenure, manager evaluation, business performance and compensation policy.

16. JDS Predictive Modelling

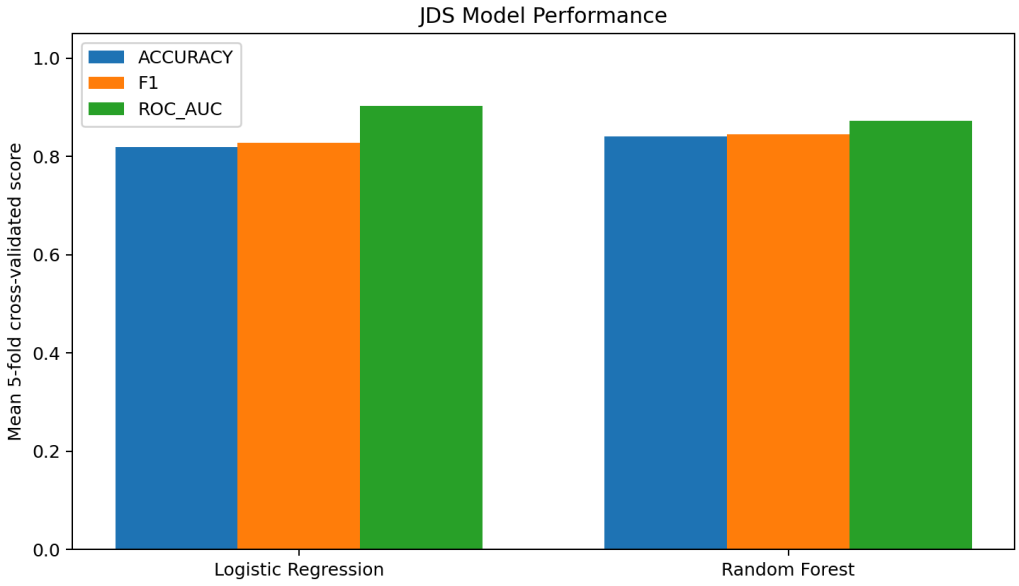


Figure 5. Five-fold cross-validated performance of JDS classification models.

Two models were evaluated: Logistic Regression as an interpretable linear baseline and Random Forest as a nonlinear ensemble model. Because the dataset contains only 139 observations, a stratified five-fold cross-validation design was used rather than relying on a single train/test split.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	81.9% ± 7.5	81.7%	84.8%	82.8%	90.3%
Random Forest	84.1% ± 7.9	84.9%	85.0%	84.5%	87.3%

Random Forest provides the stronger classification accuracy and F1 in this analysis, while Logistic Regression produces the higher mean ROC-AUC. This suggests that the relationship between skill dimensions and the binary target may contain nonlinear structure, while the linear model remains useful as a transparent benchmark.

Random Forest feature importance ranks dashboard and storytelling skills highest, followed by mathematics/statistics, AI/ML, coding and big-data skills. This ranking should be treated as model-specific evidence. It does not mean dashboard/storytelling is universally more important than coding; rather, within this fitted model and this sample, it contributed more to impurity-based separation of the classes.

The recommended Phase 1 interpretation is therefore: the JDS sample contains a measurable association between stronger technical skill profiles and high salary-hike classification, with dashboard/storytelling and mathematics/statistics showing particularly strong signals.

17. Data Analysis: Senior Data Scientist Personality Traits

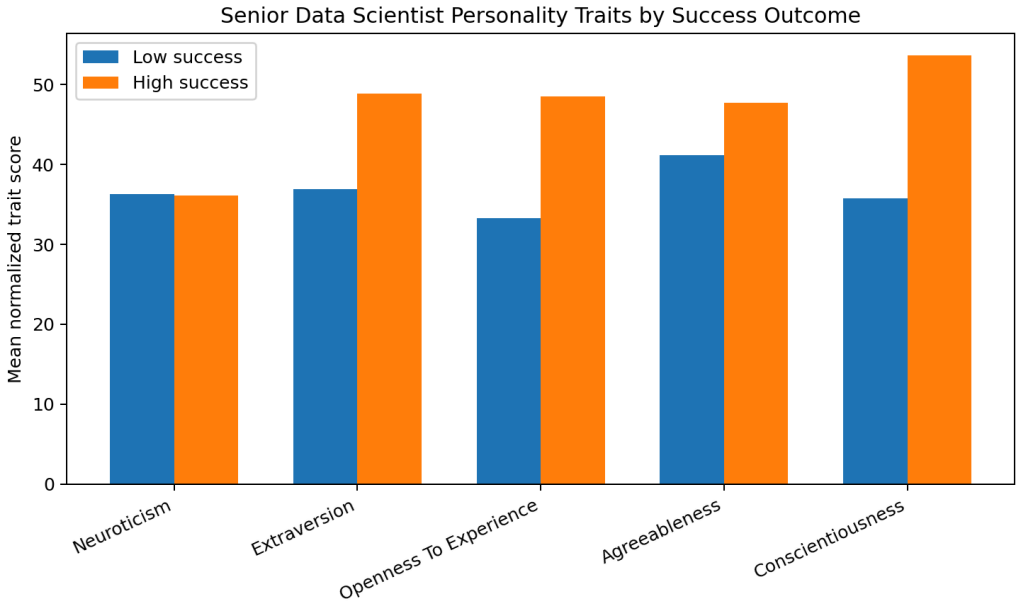


Figure 6. Mean personality-trait scores by success classification in SDS.

The SDS dataset contains 161 senior/customer-facing data scientists. The target is almost balanced, with 52.8% classified as high success and 47.2% as low success.

Trait	Low success mean	High success mean	Difference
Neuroticism	36.26	36.13	-0.13
Extraversion	36.88	48.86	11.98
Openness To Experience	33.32	48.49	15.18
Agreeableness	41.12	47.72	6.60
Conscientiousness	35.74	53.68	17.95

The strongest observed differences are in conscientiousness, openness to experience and extraversion. High-success records have mean conscientiousness of approximately 53.68 compared with 35.74 in the low-success group. Openness is approximately 48.49 versus 33.32, and extraversion approximately 48.86 versus 36.88. Agreeableness shows a smaller but still visible difference. Neuroticism is nearly identical between the two groups.

Correlation with the binary success classification is strongest for conscientiousness (approximately 0.68), openness (approximately 0.67) and extraversion (approximately 0.49). Neuroticism is essentially uncorrelated in this sample. The results are consistent with the interpretation that the observed success classification is more strongly associated with certain positive engagement, curiosity and organization-related traits than with neuroticism.

18. SDS Predictive Modelling

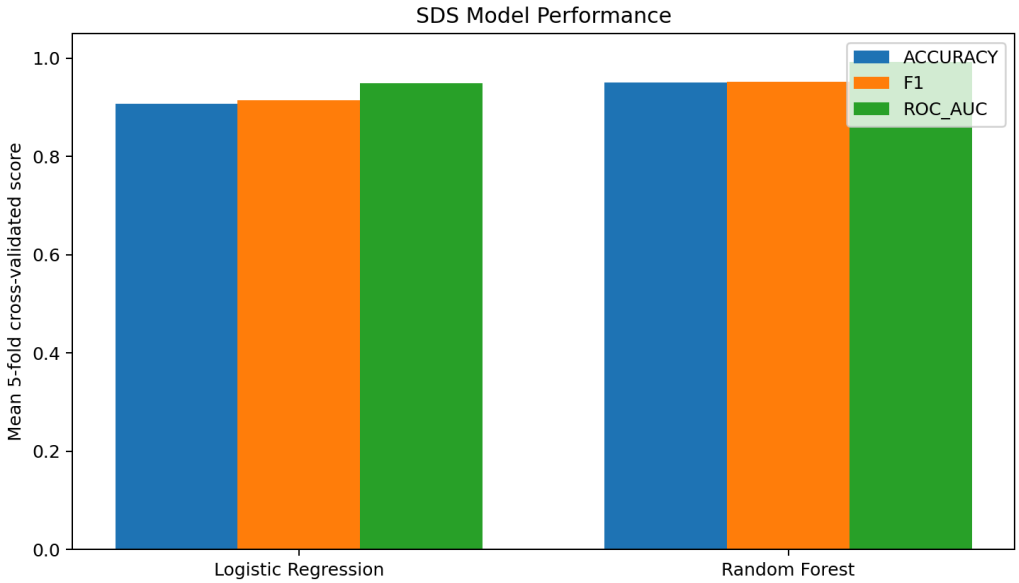


Figure 7. Five-fold cross-validated performance of SDS classification models.

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	90.7% ± 5.9	89.2%	94.1%	91.4%	94.9%
Random Forest	95.0% ± 4.2	94.3%	96.5%	95.2%	99.2%

Random Forest substantially outperforms the linear baseline on all primary metrics in the SDS sample. Its mean cross-validated accuracy is approximately 95.0%, mean F1 approximately 95.2% and mean ROC-AUC approximately 99.2%. The result is encouraging for the prototype but must be interpreted cautiously because 161 observations are a small sample and the data may be masked or self-reported.

Random Forest feature importance places conscientiousness first, followed by openness to experience, agreeableness, extraversion and neuroticism. The importance ranking is directionally consistent with the descriptive group differences. However, feature importance is not a causal effect estimate and should not be used to screen or reject individuals.

For WorkforceX, the appropriate use is as an organizational research signal: teams can investigate whether senior-role success patterns are related to selected behavioral capabilities and whether development programs should include communication, adaptability, organization or other non-technical dimensions.

19. Integrated WorkforceX Analytical Framework

The findings from the four datasets are integrated into WorkforceX through a layered decision-support workflow. The first layer describes what the market is demanding. The second layer describes which technical skill profiles are associated with junior salary-hike outcomes. The third layer describes which personality dimensions are associated with senior success classifications. The fourth layer translates these signals into workforce-development questions.

Layer	Evidence	WorkforceX question	Prototype output
Market demand	Analytics Jobs + DataScience Jobs	What capabilities and roles are visible in the market?	Demand dashboard
Technical capability	JDS	Which skill dimensions distinguish high/low hike outcomes?	Skill-outcome predictor
Senior capability	SDS	Which trait dimensions distinguish high/low success outcomes?	Trait-outcome predictor
Decision support	Integrated	What should workforce development prioritize?	Role intelligence and transition views

The current prototype screenshots demonstrate this interface architecture. The Executive Dashboard presents organization-level KPIs and workforce signals. Workforce X-Ray shows employee/role readiness in the prototype. The Skill Transformation Map represents changing tasks and emerging skills. Transition Graph shows a role-to-role analysis with matched and missing skills. Role Intelligence displays current and emerging skill requirements.

The next implementation step is to replace illustrative employee-level records and hardcoded prototype signals with API-driven calculations derived from the official datasets wherever the question is supported by those datasets. The employee transition module should be explicitly marked illustrative until an actual internal workforce dataset is available.

This separation creates a credible bridge between the hackathon data and the product concept: market analytics can be demonstrated with official data now, while internal workforce transition analytics can be demonstrated as a prototype architecture awaiting organization-specific data.

20. Results and Conclusions

The analysis provides five major conclusions.

11. The analytics job market sample is skill-diverse. SQL, Python, analytics, machine learning, data analysis, R and SAS appear frequently, while business and domain skills such as finance and business analysis are also prominent.
12. Experience and salary move together in the Analytics Jobs sample. The derived experience midpoint has an approximate 0.66 correlation with the representative salary midpoint, and the mean experience midpoint rises across salary bands.
13. The DataScience Jobs sample shows a clear role hierarchy in posting volume and compensation. Business Analyst and Data Analyst have high aggregated demand, while senior technical roles and Data Architect show higher mean recorded salary levels.
14. In the JDS sample, high salary-hike records have higher average scores across all five technical dimensions. The strongest group differences are observed for dashboard/storytelling, mathematics/statistics, coding and AI/ML.
15. In the SDS sample, high-success records show markedly higher conscientiousness, openness and extraversion, with conscientiousness and openness producing the strongest model signals.

The predictive models support the descriptive findings but do not replace them. JDS Random Forest provides the strongest classification accuracy/F1 among the tested models, while Logistic Regression remains competitive on ROC-AUC. SDS Random Forest performs very strongly across all metrics. Because the outcome samples are small, these results should be viewed as proof-of-concept model performance rather than production-grade generalization.

Overall conclusion: WorkforceX can convert fragmented workforce evidence into a connected analytical narrative. The strongest Phase 1 contribution is not a single prediction number; it is the combination of market demand, capability analysis, transparent modelling and decision-support interpretation.

21. Implications

Organizational implications: Workforce leaders can use the market analysis to identify capabilities worth monitoring and can use the JDS results to design development programs around technical dimensions that show stronger associations with positive observed outcomes. The SDS analysis suggests that senior/customer-facing success may also involve non-technical dimensions such as conscientiousness, openness and extraversion in the supplied sample.

Learning and development implications: training portfolios can be aligned with both technical and behavioral capability evidence. Rather than training every employee in every technology, an organization can prioritize skill clusters that are both visible in market demand and relevant to internal capability development.

Employee implications: the system can eventually provide transparent development pathways rather than opaque labels. An employee could see current skills, future-role requirements, missing capabilities and recommended learning steps. Any employee-facing recommendation should remain advisory and should not automatically determine employment status.

Recruitment implications: market intelligence can help organizations distinguish capabilities that can reasonably be developed internally from capabilities that may require external hiring. The current data support the market side of this decision, while an actual internal workforce dataset would be required to quantify the internal gap.

Governance implications: the models should not be used as automated hiring, promotion or termination mechanisms. Personality-based predictions in particular require careful governance because they can amplify bias or encourage inappropriate psychological screening. The correct use is exploratory organizational analytics with human oversight.

Business implications: WorkforceX can be positioned as a B2B decision-support product. A future commercial model could combine subscription analytics, enterprise integrations, customized workforce studies and learning-provider partnerships. The value proposition is better workforce planning rather than simply selling a dashboard.

22. Limitations, Risks and Phase 2 Roadmap

Dataset limitations: the official brief states that the supplied data may contain meaningless, misspelled or mistyped entries, outliers and masked or self-reported information. The job datasets are samples rather than complete labor-market censuses. JDS and SDS are small outcome datasets. These limitations restrict generalization.

Model limitations: five-fold cross-validation provides a more stable estimate than a single split, but it cannot compensate for a small or non-representative sample. Model performance can change substantially on new data. Feature importance does not imply causation.

Consolidation limitation: the four datasets cannot be directly joined at the person level. WorkforceX therefore uses an evidence-layer architecture. A future enterprise deployment needs a secure internal employee-skill dataset and a governed role/skill taxonomy.

Prototype limitation: the current UI contains illustrative employee names, transition scores and role transformations. These should be clearly labeled as demo data until backed by organization-specific records. The prototype screenshots are evidence of the interface design, not evidence of real-world prediction accuracy.

Phase 2 roadmap: improve text-based skill extraction; introduce semantic skill normalization; add explainable model outputs; validate models on a larger longitudinal dataset; create role-skill graphs; integrate internal workforce data; develop the Build/Buy/Redeploy/Automate simulator; add scenario analysis by budget and time horizon; and establish model governance and bias monitoring.

Final Phase 1 recommendation: preserve the current product concept, but ensure that every quantitative claim in the demo can be traced either to an official supplied dataset, a clearly stated derivation or an explicitly labelled illustrative prototype assumption.

Appendix A. Dataset Profile and Data Quality Tables

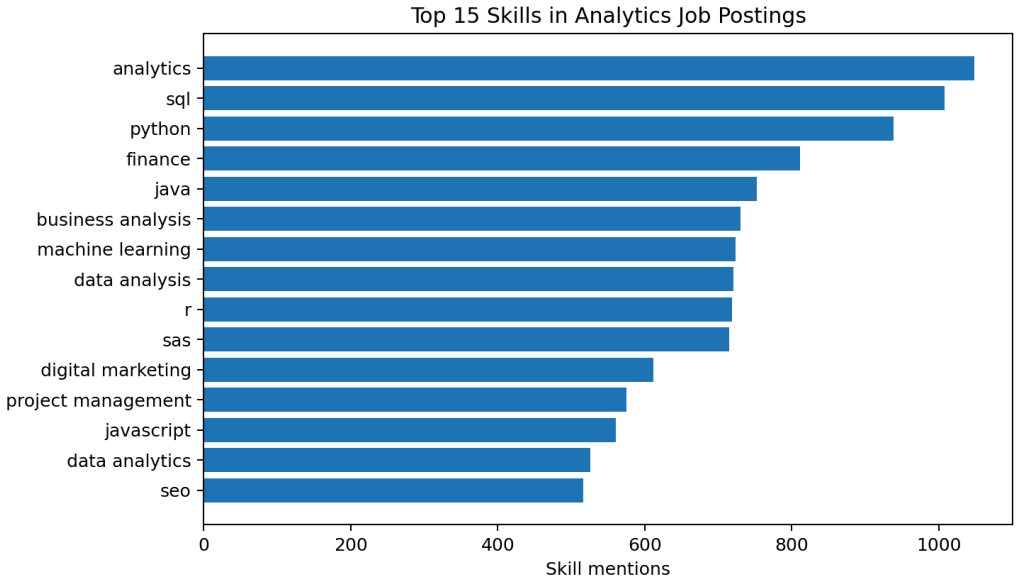
Dataset	Rows	Columns	Missing cells	Exact duplicate rows
Analytics Jobs	15,841	8	15,520	0
DataScience Jobs	1,602	8	0	0
JDS Skill Traits	139	7	0	0
SDS Personality Traits	161	7	0	0

Source: calculations from the four official hackathon-provided files. The original data dictionaries are included in the hackathon ZIP.

Analytics Jobs quality issue	Count	Share
Missing job_description	3,508	22.1%
Missing job_type	12,011	75.8%
Missing key_skills	1	0.01%
Exact duplicate rows	0	0.0%

Appendix B. Market Analysis Tables

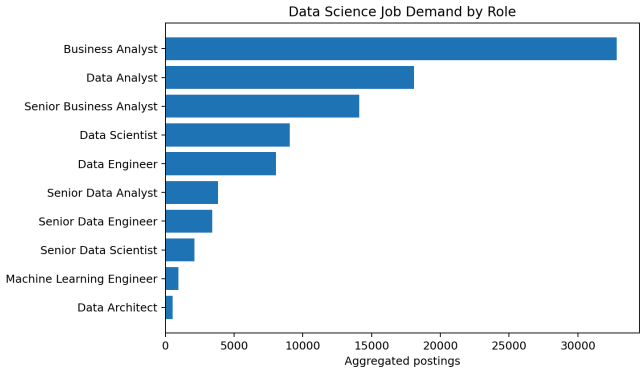
Salary band	Postings	Share
0to3	2,592	16.4%
3to6	2,239	14.1%
6to10	2,876	18.2%
10to15	3,608	22.8%
15to25	3,281	20.7%
25to50	1,245	7.9%
Top skill	**Mentions**	**% of Analytics Jobs**
analytics	1,048	6.6%
sql	1,008	6.4%
python	938	5.9%
finance	811	5.1%
java	752	4.7%
business analysis	730	4.6%
machine learning	724	4.6%
data analysis	721	4.6%
r	719	4.5%
sas	715	4.5%
digital marketing	612	3.9%
project management	575	3.6%
javascript	561	3.5%
data analytics	526	3.3%
seo	516	3.3%



Appendix Figure B1. Top skill demand.

Appendix C. Data Science Job Demand and Salary

Role	Records	Aggregated postings	Mean avg salary (Lakh)
Business Analyst	188	32,843	8.95
Data Analyst	187	18,095	5.71
Senior Business Analyst	187	14,115	13.17
Data Scientist	188	9,051	13.53
Data Engineer	188	8,044	11.81
Senior Data Analyst	187	3,825	9.57
Senior Data Engineer	183	3,411	19.00
Senior Data Scientist	185	2,129	22.29
Machine Learning Engineer	59	964	9.85
Data Architect	50	528	25.09



Appendix Figure C1. Data-science role demand.

Appendix D. JDS Statistical and Model Tables

Skill	Low-hike mean	High-hike mean	p-value
big data skills	3.750	3.940	0.195
maths-stats skills	3.830	4.712	5.71e-10
coding skills	3.853	4.644	9.49e-08
ai and ml skills	4.283	4.822	3.32e-06
dashboard and storytelling skills	3.814	4.845	3.07e-11
JDS feature		**Random Forest importance**	
dashboard and storytelling skills		0.367	
maths-stats skills		0.257	
ai and ml skills		0.160	
coding skills		0.139	

Appendix E. SDS Statistical and Model Tables

Trait	Low-success mean	High-success mean	Correlation with target
neuroticism	36.263	36.129	-0.006
extraversion	36.882	48.859	0.494
openness to experience	33.316	48.494	0.671
agreeableness	41.118	47.718	0.293
conscientiousness	35.737	53.682	0.680
SDS feature		**Random Forest importance**	
conscientiousness		0.369	
openness to experience		0.336	
agreeableness		0.142	
extraversion		0.129	
neuroticism		0.023	

Appendix F. Equations and Analytical Definitions

1. Mean: $\bar{x} = (\Sigma x_i) / n$

2. Experience midpoint: E_mid = (E_min + E_max) / 2

3. Salary-band midpoint: S_mid = (S_low + S_high) / 2

4. Skill frequency: F_s = number of normalized postings containing skill s

5. Logistic regression probability: $P(Y=1|X) = 1 / (1 + e^{-(\beta_0 + \beta_1 X_1 + ... + \beta_k X_k)})$

6. Accuracy: (TP + TN) / (TP + TN + FP + FN)

7. Precision: TP / (TP + FP)

8. Recall: TP / (TP + FN)

9. F1 score: 2 × Precision × Recall / (Precision + Recall)

10. Prototype transition score (illustrative): Transition Score = Weighted Matched Skills / Weighted Target Skills × 100. This formula is a proposed product-logic example and is not a validated prediction model.

All equations are used as analytical definitions or prototype scoring logic. They should not be interpreted as causal workforce laws.

Appendix G. WorkforceX Prototype Screenshots

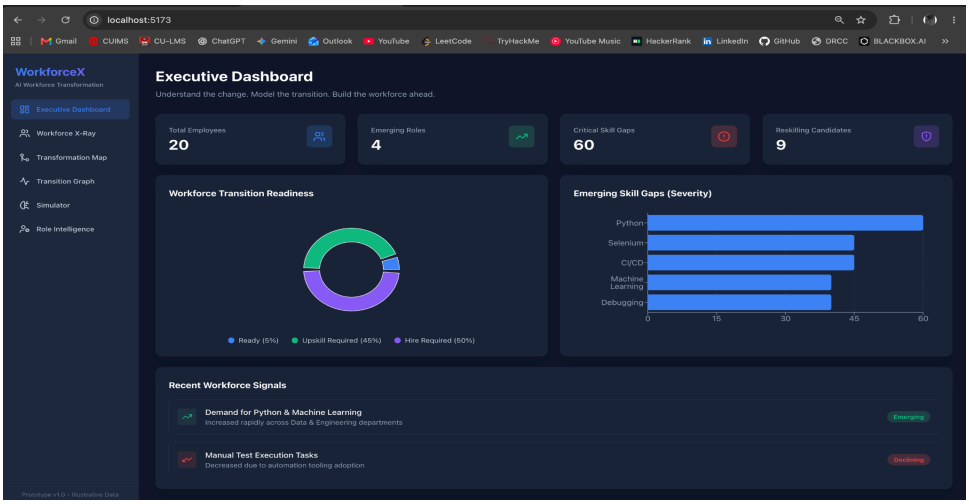


Figure G.1. Executive Dashboard.

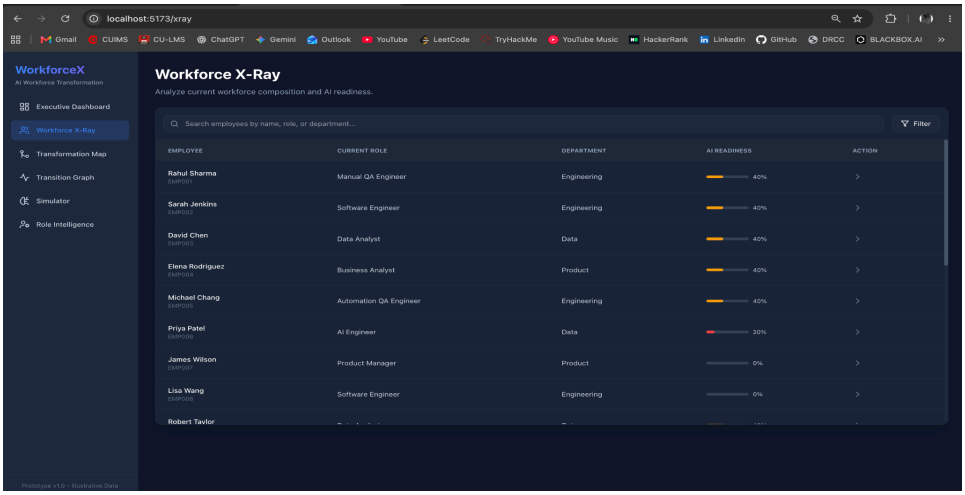


Figure G.2. Workforce X-Ray.

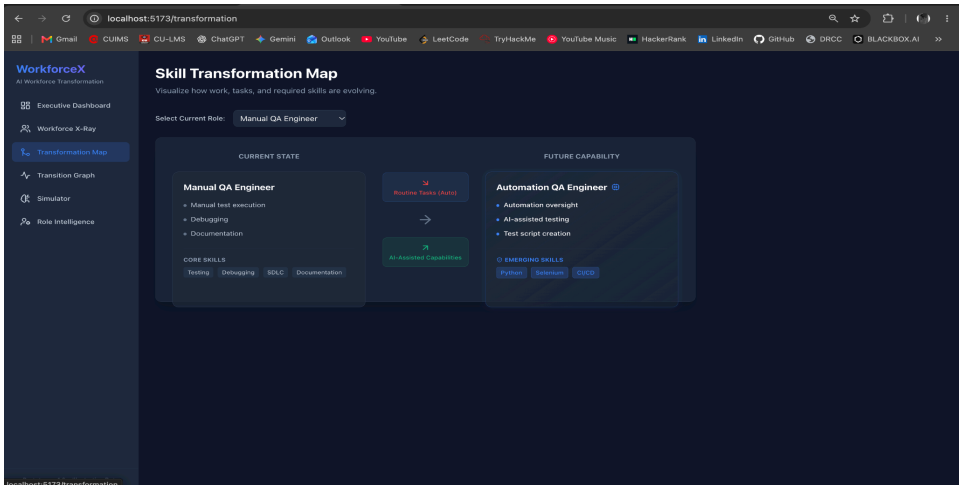


Figure G.3. Skill Transformation Map.

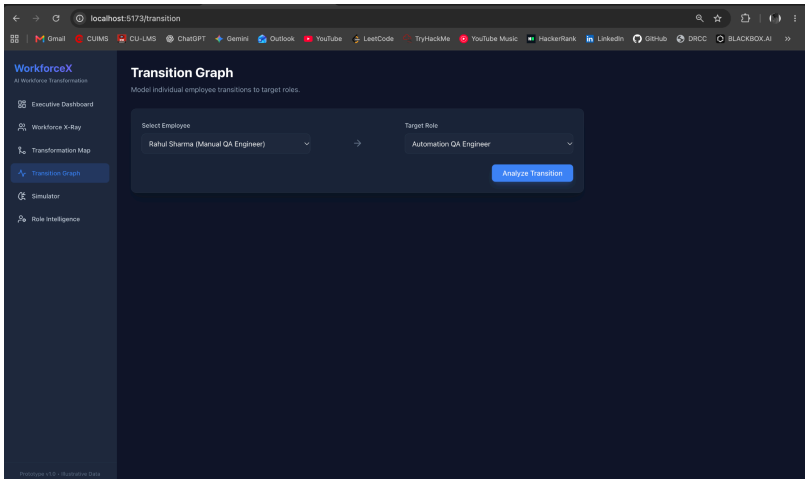


Figure G.4. Transition Graph – input screen.

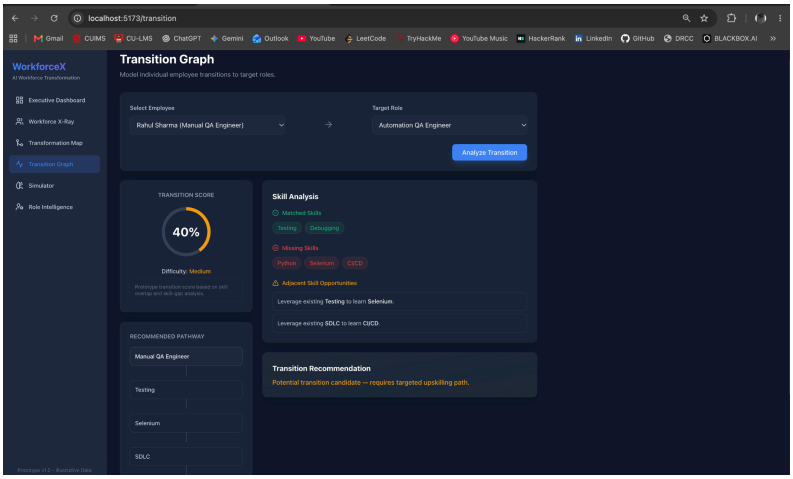


Figure G.5. Transition Graph – analyzed result.

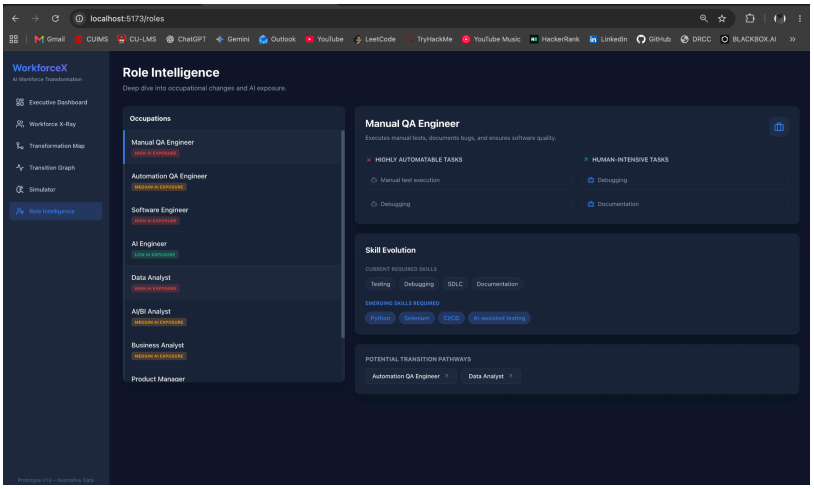


Figure G.6. Role Intelligence.

Appendix H. References and Source Materials

16. Hackathon-provided Data Description Doc.pdf — official dataset descriptions and data dictionaries.
17. Hackathon-provided Problem Context Brief, SAS VFL Demos, Guidelines Dos and Donts.pdf — official problem context, evaluation structure, approach-note requirements and do/don't guidance.
18. Analytics Jobs.csv — official hackathon-provided dataset.
19. DataScience Jobs.csv — official hackathon-provided dataset.
20. JDS Skill Traits.xlsx — official hackathon-provided dataset.
21. SDS Personality Traits.xlsx — official hackathon-provided dataset.
22. Team Mind Spark WorkforceX prototype — screenshots included in Appendix G.

Note on evidence: this report intentionally avoids external labor-market statistics because the hackathon instructions direct teams to work with the supplied datasets and the supplied files are sufficient for the Phase 1 analytical narrative. Future phases may introduce external evidence only if permitted by the organizers.

End of Phase 2 Approach Note — Team MIND SPARK.